

Online Appendix to Circular Capital in the Artificial Intelligence Stack: A Game-Theoretic Analysis of Market Power, Financial Fragility, and Welfare

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This Online Appendix accompanies the main manuscript submitted to the 7th Financial Economics Meeting (FEM-2026) and, on the fast track, to the International Journal of Finance & Economics. It contains the evidence-grading apparatus (Appendix A), the retirement-exposure appendix (Appendix B), and the assumptions, caveats, and BCR-arithmetic appendix (Appendix C).

A Confidence, Evidence Grades, and Likelihood Scale

The registry in the main text tags every claim with a likelihood and an evidence grade. This appendix defines both scales and documents what backs each tag.

How seriously to take each claim. Every numbered prediction carries two separate tags. **Likelihood** is the author’s assessed probability that the claim proves right, on the fixed scale in Table 1 — a judgment in the forecasting tradition (IPCC, Tetlock), not a model estimate. **Confidence** rates the quality of the evidence behind that judgment (High = contracts plus surviving stress tests; Medium = scenario arithmetic that moves with inputs; Low = no evidentiary basis). Model vote-shares (16/16, 100%, 65%) measure *robustness*, and formation scores (82.9% etc.) measure *vulnerability* — neither is the probability of the event. The two numbers that look most like probabilities in this paper are therefore the two you must not mistake for forecasts; the actual event probabilities are the author’s, stated in the registry.

Table 1. Likelihood scale for author-assessed probabilities. Used only in the prediction registry of the main text; nowhere else in the paper does a percent sign denote an event probability.

Wording	Assessed probability
Very likely	85–95%
Likely	65–80%
More likely than not	50–65%
Toss-up	~50%
Unlikely	below 35%

A.1 Evidence behind each likelihood

A likelihood without a stated evidence grade is advertising. Table below documents what each registry tag in the main text rests on.

High confidence requires a mechanism plus survival: the claim follows from signed contracts (not sentiment or multiples) *and* survives every stress battery in the main text. Only P2 clears both bars — the ranking follows the contract book and holds 16/16 one-at-a-time with 100% joint-draw agreement, so its ~90% is the paper’s firmest number.

Medium confidence means scenario arithmetic with stated parameters: the direction is pinned down but the level moves with inputs. P1 (~65%) rests on three simultaneously present preconditions plus live credit repricing (215bp, BBB-) — strong pattern evidence, no base rate, hence Likely rather than Very likely. P3 (~55%) inherits two layers of parameter uncertainty (impairment share, capex-cut share), so its honest expression is the \$100–150B band, not the \$139.1B point. P5 (~60%) is capped by the formation rank’s own weaker record (65% joint agreement, disclosed throughout). P6 (~60%) assumes equity buffers stay large; homogeneous GPU collateral is exactly the channel that would break it. P7 (~70%) rates the remedy *order*, which survives both the DWL and burst denominations — the levels do not deserve the same grade.

Low confidence means no evidentiary basis exists. P8 is the only resident: no timing model exists, so no probability is assigned.

Table 2. Evidence behind each registry likelihood. Vote-shares are robustness checks, not event probabilities.

ID	Likelihood rests on	What would cut it in half
P1 (~65%)	3/3 historical preconditions present; credit charging (215bp, BBB-)	Preconditions unwind without repricing
P2 (~90%)	Contract incidence; 16/16 OAT; 100% joint; seed + noise screens	Ranking flips on re-estimated exposures
P3 (~55%)	Two-channel arithmetic on BEA GDP; capex share 86%	Capex-cut share proves overstated
P4 (~70%)	Densest round-trippers score highest (82.9%)	First break lands elsewhere
P5 (~60%)	Logit drivers; circular exposure dominates (w=1.6)	Joint agreement below 50% on refresh
P6 (~60%)	DebtRank x1.032 on large equity buffers	Fire-sale channel activates
P7 (~70%)	BCR order stable across two denominations	Rank reversal in either table
P8 (none)	No timing model exists	—

B Who Holds the Wealth Channel: Retirement Exposure

The \$19.1B wealth channel is computed at a uniform 4 cents of consumption per dollar of equity loss. Two sourced facts qualify who bears it. *First, the exposure stock:* 401(k) equity funds hold \$3.7T and IRA equity funds \$4.8T ([Investment Company Institute, 2026](#)); at the Magnificent-7’s ~34% S&P 500 weight ([DataTrek via MarketWatch, 2025](#)), fund-held retirement equity carries a ~\$2.9T Big Tech slice — a floor under a market-weight assumption (hybrids, direct pension holdings, and taxable accounts excluded). *Second, the loss lands where the MPC is lowest:* the top decile holds ~87% of equities ([Federal Reserve DFA, 2025](#)), while the Fed’s heterogeneous estimates put top-quintile MPC at 0.8 cents against 7.5 cents for the bottom 80% ([Board of](#)

Governors, 2025).

Rerunning the paper’s own \$477.5B implied market-cap loss (\$19.1B / 0.04) through holder-weighted MPCs gives a \$6.1–7.0B consumption memo (Fig. 1, right panel; full arithmetic in `tables/backtest/retirement_exposure.csv`, built by `build_retirement_exposure.py` from live Table 7.12 values). The reading cuts against alarmism on the aggregate: the uniform 4-cent assumption *overstates* the consumption drag, because wealthy holders absorb most of the loss at 0.8 cents. The GDP headline stands at \$139.1B; the memo combined figure is ~\$126–127B.

The contraction risk is compositional, not larger. The \$33–48B balance-sheet slice landing on 7.5-cent households is a retirement-adequacy shock, and a consumption flow understates the welfare loss for near-retirees drawing fixed balances. Three fringe channels widen that slice beyond fund-held equity: US private credit (~\$1.7–1.8T) with listed BDCs (~\$500B) near \$0.78 per \$1.00 of NAV and retail private-credit assets up 43.7% since end-2024 (PitchBook / Morningstar, 2026); public pensions pushing into private markets (CalPERS: private equity 17%, private debt 8%) (CalPERS, 2025); and GPU-collateralized neocloud debt (CoreWeave \$14–19B, interest near a third of revenue) (TechCrunch / PitchBook, 2026) — the collateral class the top-BCR remedy caps. Executive Order 14330 (August 2025) opening 401(k)s to alternatives accelerates all three (White House, 2025).

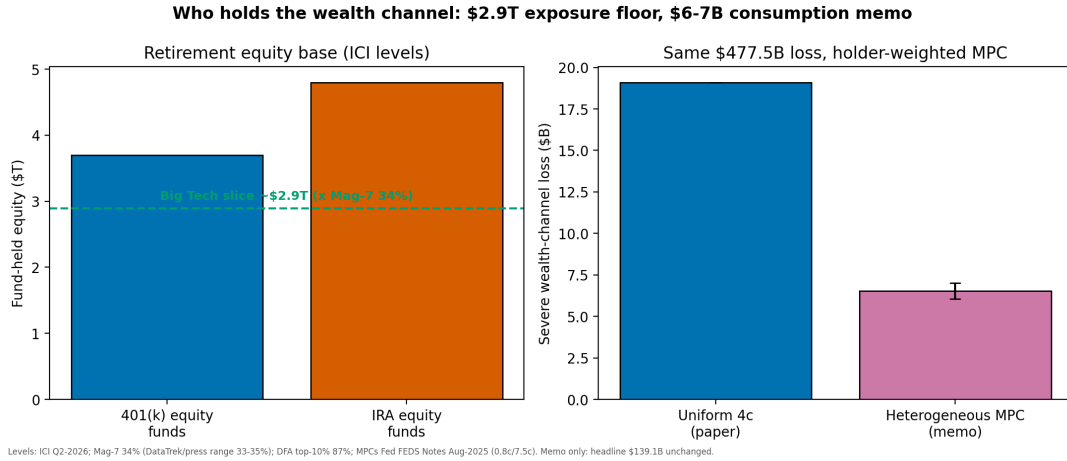


Figure 1. Left: the retirement equity base behind the wealth channel (ICI Q2-2026 levels); the dashed line is the ~\$2.9T Big Tech slice at a 34% index weight — a floor. Right: the same \$477.5B equity loss under the paper’s uniform 4-cent MPC versus holder-weighted MPCs (0.8c top quintile, 7.5c rest) — the aggregate drag shrinks while its incidence concentrates on retirement holders.

C Assumptions, Caveats, and Updating Protocol

The games are static and one-shot, missing learning and entry; the cascade is static reverberation, not dynamic clearing; the GDP math is US-calibrated without multipliers; formation odds are scenario scores, not measured probabilities. Policy BCRs are ex-ante engineering estimates with scenario-assumed reduction rates, not revealed-preference measures; their ordering is more credible than their levels, and prevention is priced, not distributional incidence. Chinese open-weight systems are excluded by the China-labs rule: the published formation odds carry no open-source margin-compression channel, which instead runs as an explicit scenario, while Chinese labs themselves contribute no edge to the frame. Geography porting exists as stated scenarios (Table 7.19) still awaiting local vintages. Treat every dollar figure as a marker and every ranking as the claim. When funding rounds reprice, tranches activate, or spreads gap, re-run the pipeline: if the burst order moves, this prediction moves with it — that updatability

is the difference between a standing warning and a stale one.

C.1 BCR arithmetic, break-even conditions, and what the ratios omit

Every benefit–cost ratio in this paper is gross benefit over cost, computed identically in both policy families and audited row by row in pipeline Table 6.4. Welfare family: $B_s = \text{DWL} \times r_s$, $N_s = B_s - C_s$, $\text{BCR}_s = B_s/C_s$ with $\text{DWL} = 416.77$ billion USD; burst family: the same form with the severe burst total (\$478.3B) as the denominator — the two families must never be compared rate-to-rate. The break-even reduction $r^* = C_{\text{base}}/\text{denominator}$ is the effectiveness level at which the base-case cost exactly pays for itself: welfare break-evens run 0.48% (antitrust, base 25%), 0.72% (data sharing, base 18%), 1.20% (interoperability, base 35%), 1.92% (subsidies, base 22.5%) and 3.60% (combined, base 55%); burst break-evens run 0.21% (lending caps, base 12%), 0.42% (disclosure, base 15%), 1.25% (reserve, base 10%) and 1.88% (combined, base 30%). Every base case clears its break-even by an order of magnitude or more, and every welfare conservative case still pays ($\text{BCR} \geq 1$, lowest 5.2 for subsidies) — so the *ranking* survives broad sensitivity even where the point levels should not travel. What the ratios omit is stated, not hidden: costs and benefits are single-year and undiscounted (no horizon or discount rate is estimated); incidence across taxpayers, workers and users is outside the frame; implementation delay and evasion are unmodeled; private-sector compliance costs, administrative burdens and dynamic investment-incentive effects sit on the cost side only as the narrow agency budgets in Table 6.4, so every BCR is gross of the regulatory burden it would impose — again favoring the ranking over the level. Burst-family reduction rates are scenario assumptions with mechanism analogies, not literature estimates.

Regulatory lift. Disclosure should follow this paper’s booking rule — committed face values reported, conditional tranches disclosed by trigger — which regulators can lift verbatim into reporting templates. Competition screening should precede price policing (within-layer HHIs 4,057–5,632, all highly concentrated; cross-stack top-two share 86.2%): review consolidation and exclusivity clauses in the booked deal graph before benchmarking cloud margins. The European Union offers the natural experiment for the interoperability remedy: DMA enforcement data on portability mandates can calibrate the reduction rates behind the 29.2 BCR, replacing engineering assumptions with revealed compliance effects. Finally, keep the ex-post shelf stocked — public-cloud conversion designs, worker protections, structural-reform statutes — keyed to the severe-burst trigger (\$139.1B GDP cost) this paper quantifies.

References

All citations below follow APA style (6th edition) as required by the journal. Every entry is cited in this appendix; every in-text citation resolves below.

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